Supplementary material to the Corrigendum to "An accelerated Benders decomposition algorithm for the solution of the multi-trip time-dependent vehicle routing problem with time windows" [European Journal of Operational Research 317 (2024) 500–514]: corrected Tables 2–5 and Fig. 4(a)

Table 2 (corrected): Computational Results for Instances with 15 Customers, 5 Speed Intervals and 2 Vehicles (T15-5-2)

INSTANCES		CPU TIME (secs)						OBJECTIVE FUNCTION VALUE						OPTIMALITY GAP						
A/A	CASE	MODEL-1	MODEL-2	MODEL-1-VI	MODEL-2-VI	BENDERS-DSEC-VI-SMS	BENDERS-DSEC-VI-SMS-MC	MODEL-1	MODEL-2	MODEL-1-VI	MODEL-2-VI	BENDERS-DSEC-VI-SMS	BENDERS-DSEC-VI-SMS-MC	MODEL-1	MODEL-2	MODEL-1-VI	MODEL-2-VI	BENDERS-DSEC-VI-SMS	BENDERS-DSEC-VI-SMS-MC	ALNS-VND [Pan et al. (2021)]
1	c201	**1.5**	**0.7**	**1.4**	**1.0**	**0.9**	**1.3**	211.75	211.75	211.75	211.75	211.75	211.75	0%	0%	0%	0%	0%	0%	0%
2	c202	3600	3600	**341.3**	**92.7**	**1168**	**55.4**	203.07	203.07	203.07	203.07	203.07	203.07	7%	**15%**	0%	0%	0%	0%	0%
3	c203	3600	3600	**2148**	**571.0**	**2150**	**85.4**	203.07	**214.19**	203.07	203.07	203.07	203.07	**17%**	**25%**	0%	0%	0%	0%	0%
4	c204	3600	3600	**3546**	**813.9**	**2349**	**473.5**	188.68	188.68	188.68	188.68	188.68	188.68	**17%**	20%	0%	0%	0%	0%	0%
5	c205	**14.5**	**11.6**	**8.2**	**3.6**	**3.7**	**1.7**	208.79	208.79	208.79	208.79	208.79	208.79	0%	0%	0%	0%	0%	0%	0%
6	c206	**30.3**	**31.3**	**34.3**	**3.5**	**5.7**	**4.6**	208.79	208.79	208.79	208.79	208.79	208.79	0%	0%	0%	0%	0%	0%	0%
7	c207	**839.8**	**1213**	**221.0**	**44.2**	**1007**	**30.0**	204.85	204.85	204.85	204.85	204.85	204.85	0%	0%	0%	0%	0%	0%	0%
8	c208	**578.0**	**1783**	**373.4**	**82.0**	**238.4**	**15.9**	203.07	203.07	203.07	203.07	203.07	203.07	0%	0%	0%	0%	0%	0%	0%
9	r201	**23.6**	**25.6**	**19.5**	**10.2**	**302.3**	**207.8**	**343.68**	**343.68**	**343.68**	**343.68**	**343.68**	**343.68**	0%	0%	0%	0%	0%	0%	0%
10	r202	3600	3600	**3600**	**3526**	3600	3600	**304.67**	**297.80**	**297.80**	**297.80**	–	**324.13**	**11%**	**12%**	**7%**	0%	100%	**9%**	0%
11	r203	3600	3600	3600	**3600**	3600	3600	**307.08**	**315.53**	**309.25**	**297.80**	–	–	**12%**	**22%**	**13%**	**9%**	100%	100%	**3%**
12	r204	3600	3600	3600	3600	3600	3600	**284.92**	**283.61**	**289.22**	**277.22**	–	**–**	**14%**	**20%**	**18%**	**14%**	100%	**100%**	**3%**
13	r205	**39.3**	**16.4**	**31.5**	**5.5**	**18.3**	**9.6**	278.08	278.08	278.08	278.08	278.08	278.08	0%	0%	0%	0%	0%	0%	0%
14	r206	**225.5**	**340.7**	**261.1**	**82.9**	**87.2**	**31.2**	248.91	248.91	248.91	248.91	248.91	248.91	0%	0%	0%	0%	0%	0%	0%
15	r207	**986.7**	**2053**	**950.1**	**298.5**	**111.9**	**62.1**	248.91	248.91	248.91	248.91	248.91	248.91	0%	0%	0%	0%	0%	0%	0%
16	r208	**3600**	**3600**	**593.4**	**544.6**	**213.8**	**53.5**	238.23	238.23	238.23	238.23	238.23	238.23	**1%**	**6%**	0%	0%	0%	0%	0%
17	r209	**90.4**	**40.2**	**94.8**	**27.8**	**15.1**	**5.3**	263.48	263.48	263.48	263.48	263.48	263.48	0%	0%	0%	0%	0%	0%	0%
18	r210	**2724**	3600	**3600**	**3481**	**3389**	**1420**	292.15	292.15	292.15	292.15	**292.15**	292.15	0%	**11%**	**4%**	0%	**0%**	0%	0%
19	r211	**962.4**	**316.9**	**898.7**	**241.7**	**554.7**	**232.1**	247.23	247.23	247.23	247.23	247.23	247.23	0%	0%	0%	0%	0%	1%	0%
20	rc201	**48.8**	**45.2**	**29.8**	**17.2**	**529.1**	**341.6**	**313.08**	**313.08**	**313.08**	**313.08**	**313.08**	**313.08**	0%	0%	0%	0%	0%	0%	0%
21	rc202	3600	3600	3600	3600	3600	3600	235.37	235.37	235.37	235.37	–	–	**26%**	**29%**	12%	14%	100%	100%	**2%**
22	rc203	3600	3600	3600	3600	3600	3600	198.64	198.64	198.64	198.64	–	–	**28%**	32%	**28%**	**29%**	100%	100%	**13%**
23	rc204	3600	3600	3600	3600	3600	3600	192.18	**231.33**	192.18	192.18	–	–	32%	**45%**	**30%**	**32%**	100%	100%	**14%**
24	rc205	**1745**	**878.4**	**316.5**	**219.0**	**3600**	**3600**	**236.69**	**236.69**	**236.69**	**236.69**	**–**	**–**	0%	0%	0%	0%	**100%**	**100%**	0%
25	rc206	**318.8**	**115.8**	**235.2**	**129.1**	**84.6**	**12.0**	204.22	204.22	204.22	204.22	204.22	204.22	0%	0%	0%	0%	0%	0%	0%
26	rc207	**3600**	3600	3600	**3364**	3600	3600	189.06	189.06	189.06	189.06	–	–	**14%**	23%	**26%**	0%	100%	100%	0%
27	rc208	3600	3600	3600	3600	3600	**757.0**	**157.01**	**157.01**	**157.01**	**157.01**	–	**157.01**	**15%**	**18%**	**16%**	**17%**	100%	**0%**	**0%**

Bold: value different from the published table. Objective values are given with two decimals and compared with the published (integer) values after rounding. CPU times in seconds; 3600 = time limit of 1 hour reached; CPU times reached at the time limit are not marked bold. –: no feasible solution found within the time limit (optimality gap 100%). Benders runs that exceeded the time limit are reported at 3600 s; none of them found a feasible solution after 3600 s. The Benders variants stop when the Benders optimality gap is below ε = 1%. ALNS-VND: optimality gap of the best ALNS-VND solution (C_bst) reported by Pan, Zhang, & Lim (2021, Table 4), recomputed with the best lower bound obtained in this study. Computations: IBM ILOG CPLEX 22.1.1; Intel(R) Core(TM) i7-4790 CPU @ 3.6 GHz, 32 GB RAM.

Table 3 (corrected): Computational Results for Instances with 15 Customers, 11 Speed Intervals and 2 Vehicles (T15-11-2)

INSTANCES		CPU TIME (secs)						OBJECTIVE FUNCTION VALUE						OPTIMALITY GAP						
A/A	CASE	MODEL-1	MODEL-2	MODEL-1-VI	MODEL-2-VI	BENDERS-DSEC-VI-SMS	BENDERS-DSEC-VI-SMS-MC	MODEL-1	MODEL-2	MODEL-1-VI	MODEL-2-VI	BENDERS-DSEC-VI-SMS	BENDERS-DSEC-VI-SMS-MC	MODEL-1	MODEL-2	MODEL-1-VI	MODEL-2-VI	BENDERS-DSEC-VI-SMS	BENDERS-DSEC-VI-SMS-MC	ALNS-VND [Pan et al. (2021)]
1	c201	**5.7**	2.1	**2.9**	**2.4**	**4.0**	**1.9**	211.75	211.75	211.75	211.75	211.75	211.75	0%	0%	0%	0%	0%	0%	0%
2	c202	3600	3600	**3600**	**3600**	**3600**	**376.8**	206.14	203.07	203.07	203.07	–	203.07	18%	**25%**	**14%**	**12%**	**100%**	0%	0%
3	c203	3600	3600	**3600**	**3600**	3600	**452.3**	**232.66**	**203.07**	203.07	203.07	–	203.07	**31%**	**28%**	**14%**	**15%**	**100%**	0%	0%
4	c204	3600	3600	**3600**	**3600**	**3600**	**1203**	–	–	188.68	188.68	–	188.68	100%	**100%**	**11%**	**11%**	**100%**	0%	0%
5	c205	**76.2**	**243.9**	**27.2**	**19.1**	**18.9**	**4.0**	208.79	208.79	208.79	208.79	208.79	208.79	0%	0%	0%	0%	0%	0%	0%
6	c206	**817.9**	**2127**	**358.0**	**169.2**	**116.9**	**7.4**	208.79	208.79	208.79	208.79	208.79	208.79	0%	0%	0%	0%	0%	0%	0%
7	c207	3600	3600	**536.1**	**335.6**	**1973**	**45.7**	204.85	204.85	204.85	204.85	204.85	204.85	**8%**	**9%**	0%	0%	0%	0%	0%
8	c208	3600	3600	**3600**	**2447**	**978.0**	**44.5**	203.07	203.07	203.07	203.07	203.07	203.07	**21%**	**23%**	**10%**	0%	0%	0%	0%
9	r201	**41.2**	**69.7**	**65.7**	**24.0**	**122.2**	**115.0**	341.03	341.03	341.03	**341.03**	341.03	341.03	0%	0%	0%	0%	0%	0%	0%
10	r202	3600	3600	3600	3600	3600	**3447**	**316.27**	**297.80**	**311.66**	**297.80**	–	**297.52**	**19%**	**24%**	**18%**	**13%**	100%	0%	**0%**
11	r203	3600	3600	3600	3600	3600	3600	**334.95**	**–**	**316.76**	**297.52**	–	**297.52**	**23%**	**100%**	**21%**	**21%**	100%	**3%**	2%
12	r204	3600	3600	3600	3600	3600	3600	**406.74**	–	**416.17**	**276.34**	–	266.59	**44%**	100%	**47%**	**19%**	100%	**6%**	**6%**
13	r205	**34.2**	**146.3**	**58.3**	**35.8**	**58.5**	**25.7**	278.08	278.08	278.08	278.08	278.08	278.08	0%	0%	0%	0%	0%	0%	0%
14	r206	3600	3600	**3600**	**3600**	**3600**	**259.2**	248.91	**–**	**260.05**	248.91	**–**	248.91	**11%**	**100%**	**8%**	**6%**	**100%**	0%	0%
15	r207	3600	3600	3600	3600	**3600**	**483.1**	**–**	**–**	**267.05**	**260.05**	**–**	248.91	**100%**	**100%**	**15%**	**10%**	**100%**	0%	0%
16	r208	3600	3600	3600	3600	**3600**	**1086**	**–**	–	**–**	**255.59**	**–**	238.23	**100%**	100%	**100%**	**16%**	**100%**	0%	0%
17	r209	**452.9**	**3600**	**370.7**	**83.8**	**340.5**	**12.0**	263.48	263.48	263.48	263.48	263.48	263.48	0%	**5%**	0%	0%	0%	0%	0%
18	r210	3600	3600	3600	3600	**3600**	**3224**	**318.79**	**310.39**	**297.52**	**304.72**	**–**	292.15	**18%**	**26%**	**12%**	**18%**	**100%**	0%	0%
19	r211	3600	3600	**3600**	**3600**	**3600**	**3600**	**–**	**273.69**	**275.75**	**262.14**	**–**	**–**	**100%**	**21%**	**19%**	**15%**	**100%**	**100%**	**6%**
20	rc201	**107.3**	**124.2**	**28.8**	**20.6**	**18.1**	**9.9**	**288.84**	**288.84**	**288.84**	**288.84**	**288.84**	**288.84**	0%	0%	0%	0%	0%	0%	0%
21	rc202	3600	3600	3600	3600	3600	3600	**258.80**	**258.80**	235.37	235.37	–	235.37	**45%**	**48%**	**18%**	17%	100%	15%	15%
22	rc203	3600	3600	3600	3600	3600	3600	**198.64**	**198.64**	**200.70**	198.64	–	198.64	**32%**	**34%**	**31%**	30%	100%	27%	27%
23	rc204	3600	3600	3600	3600	3600	3600	**–**	–	**192.93**	192.18	–	**192.93**	**100%**	100%	**33%**	**32%**	100%	**29%**	**29%**
24	rc205	**639.3**	**3600**	**473.5**	**169.2**	**195.9**	**19.1**	**200.76**	**200.76**	**200.76**	200.76	**200.76**	**200.76**	0%	**25%**	0%	0%	0%	0%	0%
25	rc206	**485.5**	3600	**2884**	**1064**	**555.7**	**36.6**	204.22	204.22	204.22	204.22	204.22	204.22	0%	2%	0%	0%	0%	0%	0%
26	rc207	3600	3600	3600	3600	3600	3600	189.06	189.06	189.06	189.06	–	189.06	**32%**	**33%**	**29%**	28%	100%	**25%**	**25%**
27	rc208	3600	3600	3600	3600	3600	3600	**–**	**–**	**161.51**	**158.53**	–	**–**	**100%**	**100%**	**25%**	**25%**	100%	**100%**	**3%**

Bold: value different from the published table. Objective values are given with two decimals and compared with the published (integer) values after rounding. CPU times in seconds; 3600 = time limit of 1 hour reached; CPU times reached at the time limit are not marked bold. –: no feasible solution found within the time limit (optimality gap 100%). Benders runs that exceeded the time limit are reported at 3600 s; none of them found a feasible solution after 3600 s. The Benders variants stop when the Benders optimality gap is below ε = 1%. ALNS-VND: optimality gap of the best ALNS-VND solution (C_bst) reported by Pan, Zhang, & Lim (2021, Table 4), recomputed with the best lower bound obtained in this study. Computations: IBM ILOG CPLEX 22.1.1; Intel(R) Core(TM) i7-4790 CPU @ 3.6 GHz, 32 GB RAM.

Table 4 (corrected): Comparative Results of Instances Solved To Optimality And The Average CPU Time

INSTANCES			No OF INSTANCES SOLVED OPTIMALLY						AVERAGE CPU TIME (sec) FOR INSTANCES SOLVED OPTIMALLY						COMPARISON OF AVERAGE CPU TIME (sec) FOR INSTANCES SOLVED OPTIMALLY					
A/A	CUSTOMERS - SPEED INTERVALS - VEHICLES	CASE - No OF INSTANCES	MODEL-1	MODEL-2	MODEL-1-VI	MODEL-2-VI	BENDERS-DSEC-VI-SMS	BENDERS-DSEC-VI-SMS-MC	MODEL-1	MODEL-2	MODEL-1-VI	MODEL-2-VI	BENDERS-DSEC-VI-SMS	BENDERS-DSEC-VI-SMS-MC	MODEL-2 vs MODEL-1	MODEL-1-VI vs MODEL-1	MODEL-2-VI vs MODEL-2	BENDERS-DSEC-VI-SMS vs MODEL-1	BENDERS-DSEC-VI-SMS-MC vs MODEL-1	BENDERS-DSEC-VI-SMS-MC vs BENDERS-DSEC-VI-SMS
1	15-5-2	C2 - 8	5	5	8	8	8	8	**293**	**608**	**834**	**201**	**865**	**83**	**108%**	**185%**	**-67%**	**195%**	**-71%**	**-90%**
2		R2 -11	**7**	**6**	**7**	**9**	**8**	8	**722**	**465**	**407**	**913**	**587**	**253**	**-36%**	**-44%**	**96%**	**-19%**	**-65%**	**-57%**
3		RC2 - 8	**3**	3	3	4	**2**	3	**704**	**346**	**194**	**932**	**307**	**370**	**-51%**	**-72%**	**169%**	**-56%**	**-47%**	**21%**
4	15-11-2	C2 - 8	3	3	**4**	**5**	**5**	8	**300**	**791**	**231**	**595**	**618**	**267**	**164%**	**-23%**	**-25%**	**106%**	**-11%**	**-57%**
5		R2 - 11	3	**2**	**3**	**3**	**3**	**8**	**176**	**108**	**165**	**48**	**174**	**1082**	**-39%**	**-6%**	**-56%**	**-1%**	**514%**	**523%**
6		RC2 - 8	3	**1**	3	3	3	3	**411**	**124**	**1129**	**418**	**257**	**22**	**-70%**	**175%**	**237%**	**-38%**	**-95%**	**-91%**
TOTAL AVERAGE		54	**24**	**20**	**28**	**32**	**29**	**38**	**470**	**479**	**532**	**560**	**573**	**386**	**2%**	**13%**	**17%**	**22%**	**-18%**	**-33%**

Bold: value different from the published table. Solved optimally: proven optimal; for the Benders variants, Benders optimality gap below ε = 1%.
Comparison columns: relative difference of the average CPU times.

Table 5 (corrected): Comparative Results of Average Optimality Gap

INSTANCES			AVERAGE OPTIMALITY GAP							COMPARISON OF AVERAGE OPTIMALITY GAP						
A/A	CUSTOMERS - SPEED INTERVALS - VEHICLES	CASE - No OF INSTANCES	MODEL-1	MODEL-2	MODEL-1-VI	MODEL-2-VI	BENDERS-DSEC-VI-SMS	BENDERS-DSEC-VI-SMS-MC	ALNS-VND [(Pan, Zhang, & Lim, 2021)]	MODEL-2 vs MODEL-1	MODEL-1-VI vs MODEL-1	MODEL-2-VI vs MODEL-2	BENDERS-DSEC-VI-SMS vs MODEL-1	BENDERS-DSEC-VI-SMS-MC vs MODEL-1	BENDERS-DSEC-VI-SMS-MC vs BENDERS-DSEC-VI-SMS	ALNS-VND vs BENDERS-DSEC-VI-SMS-MC
1	15-5-2	C2 - 8	5%	7%	0%	0%	0%	0%	0%	**2%**	-5%	-7%	-5%	-5%	0%	0%
2		R2 -11	**3%**	**6%**	**4%**	**2%**	**27%**	**19%**	**1%**	**3%**	0%	-4%	**24%**	**16%**	**-8%**	**-19%**
3		RC2 - 8	**14%**	**18%**	**14%**	**11%**	**75%**	62%	**4%**	**4%**	**0%**	-7%	**61%**	**48%**	**-12%**	**-59%**
4	15-11-2	C2 - 8	**22%**	**23%**	**6%**	**5%**	**38%**	0%	0%	**1%**	**-16%**	**-18%**	**15%**	**-22%**	**-37%**	0%
5		R2 - 11	**38%**	**52%**	**22%**	**11%**	**73%**	**10%**	1%	**15%**	**-16%**	**-42%**	**35%**	**-28%**	**-63%**	**-9%**
6		RC2 - 8	**39%**	**43%**	**17%**	**17%**	62%	**25%**	**12%**	**4%**	**-22%**	**-26%**	**24%**	**-14%**	**-38%**	**-12%**
AVERAGE		54	**20.3%**	**25.5%**	**10.7%**	**7.4%**	**46.3%**	**18.8%**	**2.7%**	**5.2%**	**-9.6%**	**-18.1%**	**26.0%**	**-1.5%**	**-27.5%**	**-16.1%**

Bold: value different from the published table. Solved optimally: proven optimal; for the Benders variants, Benders optimality gap below ε = 1%.
Optimality gap of a run without a feasible solution = 100%. Comparison columns: difference of the average optimality gaps (percentage points).
ALNS-VND: average of the recomputed ALNS-VND gaps of Tables 2 and 3.

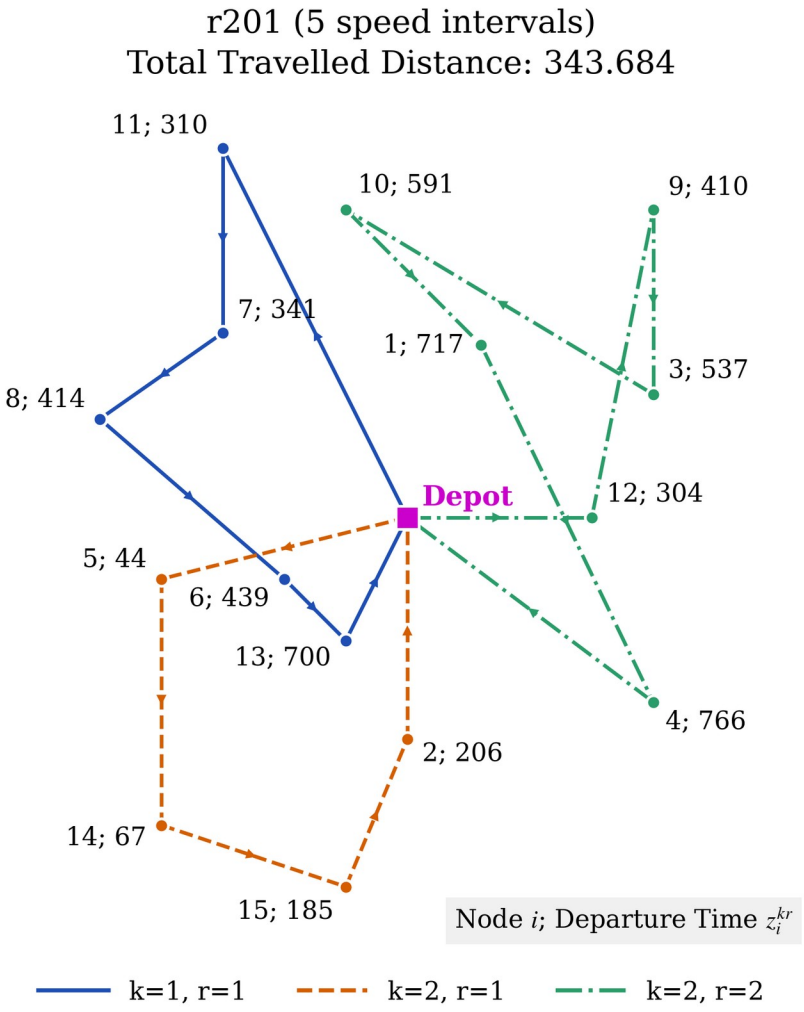


Fig. 4(a) (corrected). Optimal solution of the MTTDVRPTW for the instance r201 with 5 speed intervals (total travelled distance 343.684). Vehicle 1 performs one trip and vehicle 2 performs two trips. Fig. 4(b) (11 speed intervals) is unchanged.